What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

Alkylating Agents

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Adapted from: Alkylating and Platinum Agents

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Learning Objectives (2) Versions

After completing this brick, you will be able to:

1. Explain the mechanism of action of commonly used alkylating agents.
2. Explain the adverse effects of alkylating agents and strategies for mitigating these effects.

CASE CONNECTION

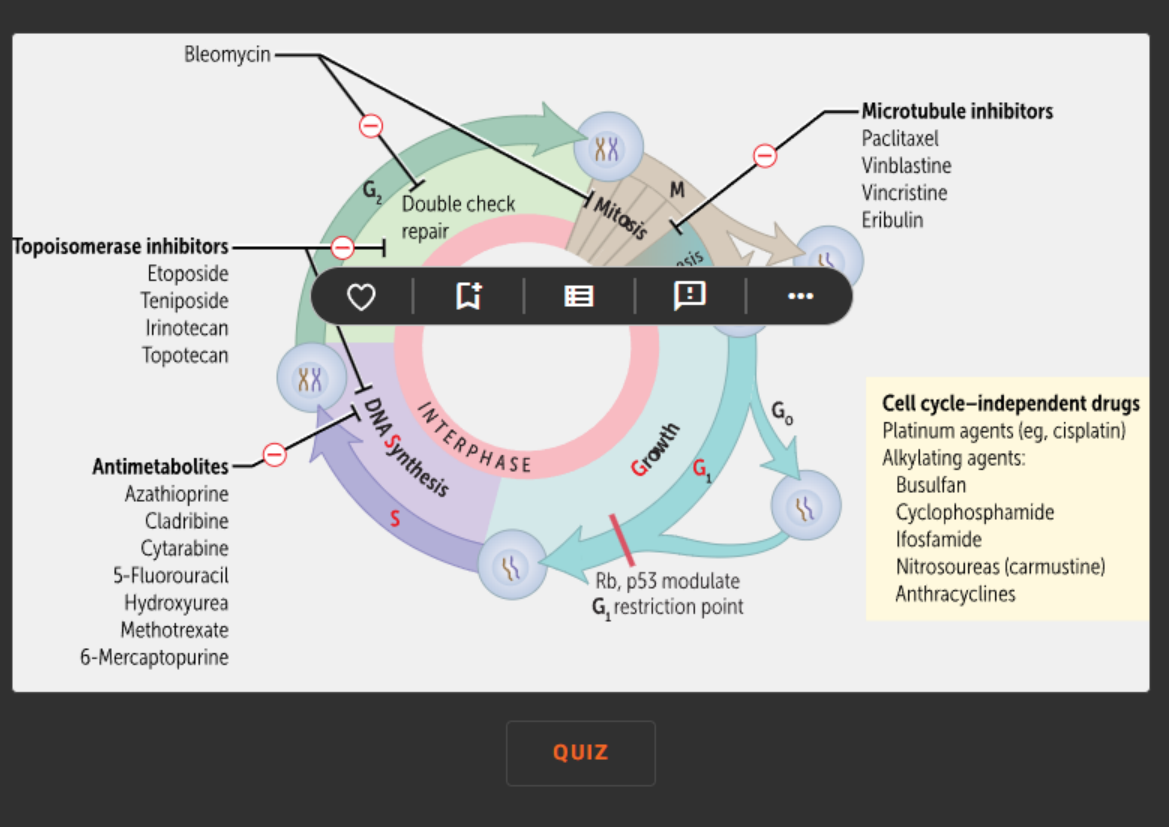
MG is a 39-year-old woman who was diagnosed with acute lymphoblastic leukemia and is considering her treatment options, and her oncologist recommended cyclophosphamide as one of the treatment options. You are a medical student on your clerkship rotations, and she asks you for your advice. "I read that cyclophosphamide is painful and can cause bleeding. I also read that it can cause hair loss. Frankly, it sounds horrible."

What will you tell MG about why therapies such as cyclophosphamide can cause such effects? What strategies can mitigate the adverse effects? Consider your answer as you read, and we'll revisit at the end of the brick.

GO TO CONCLUSION ↓

What Are Alkylating Agents? (LO#1)

Many chemotherapeutic agents exert their main effect on one phase of the cell cycle (Figure 1). By contrast, the alkylating agents and platinum agents inhibit DNA replication at all stages of the cell cycle, so they are **cell cycle independent**.



QUIZ

Figure 1

The five traditional types of alkylating and platinum agents are as follows:

- Nitrogen mustards (eg, cyclophosphamide)
- Alkyl sulfonates (eg, busulfan)
- Nitrosoureas (eg, carmustine)
- Nonclassic alkylating agents (eg, procarbazine)
- Platinum agents (eg, carboplatin, cisplatin)

What are alkylating agents?

These agents are cytotoxic because they can transfer their alkyl group to the DNA. The alkylation can occur in single or both DNA strands through cross-linking. Since cross-linking can occur with any DNA, these agents can affect both cycling and resting cells.

How do alkylating

Alkylating agents were among the first drug classes to successfully treat cancer. They add alkyl groups (carbon chains) to strands of DNA (Figure 2, left). These extra, awkward carbon strands interfere with DNA replication by **cross-linking** the DNA strand (at the guanine base) both to itself (intrastrand) and to the other DNA strand (interstrand, Figure 2, center). The cross-linked DNA then breaks, and replication is suppressed (Figure 2, right). Since cross-linking can occur with any DNA, these agents can affect both cycling and resting cells.

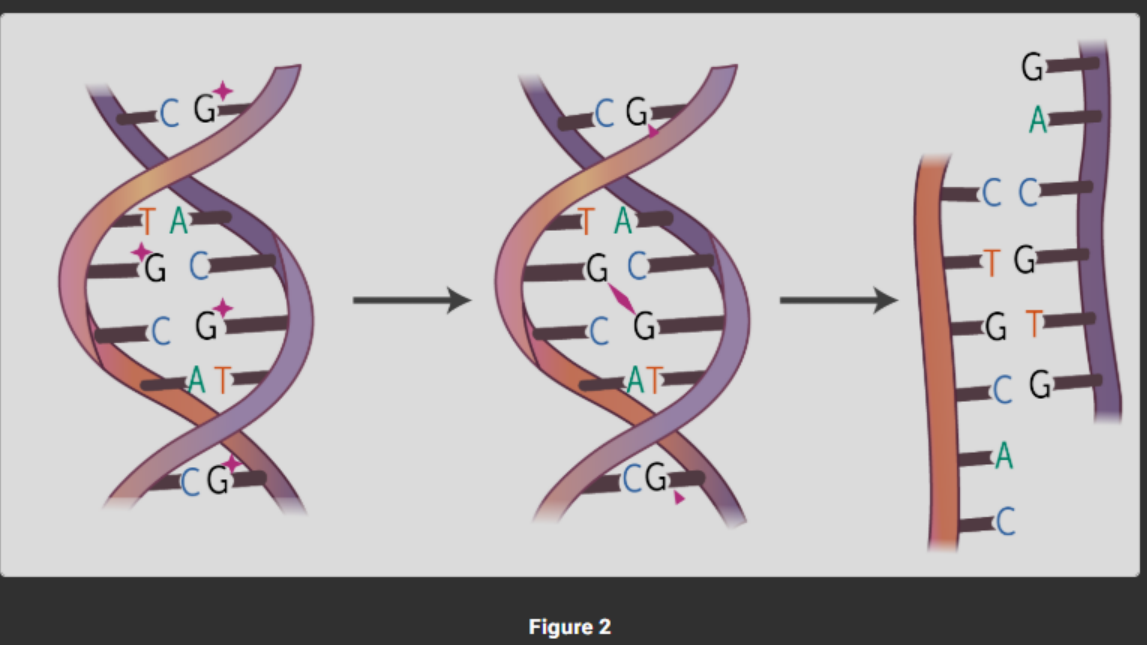


Figure 2

Differentiating intrastrand and interstrand cross-linking is important because these processes cause varying degrees of cell damage. Intrastrand cross-links can be more easily rep making them less toxic. Interstrand cros complicated homologous recombination pathway. This involves formation of double-stranded breaks, making repair more difficult and the process more toxic.

As noted, nitrogen mustards cross-link DNA strands. Drug metabolites alkylate the N-7 position of a guanosine nucleotide. Alkylating agents can covalently attach to two separate nucleotides and thus cross-link DNA (Figure 3). In this state, DNA replication cannot continue, the cell cycle is arrested, and the cell undergoes apoptosis.

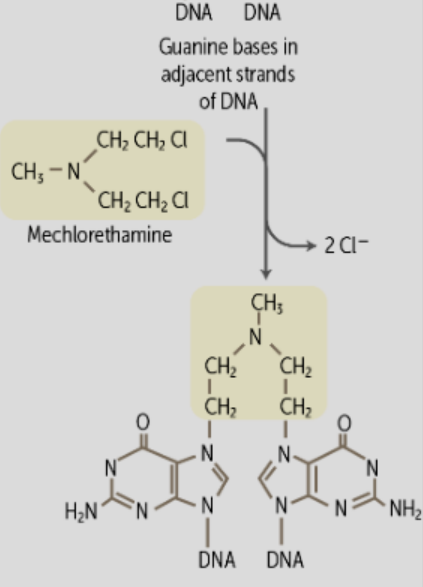

DNA DNA
Guanine bases in adjacent strands of DNA
Mechlorethamine
2 Cl−
DNA DNA
Crosslinked strands of DNA

Figure 3

What is the primary mechanism of action of alkylating agents?

CLINICAL CORRELATION

Nitrogen mustards were initially developed for use in chemical warfare during World War I to burn skin and mucous membranes.

What are some commonly used alkylating agents?

Examples of alkylating agents are: *Cyclophosphamide, Ifosfamide, Carmustine, Busulfan*

Cyclophosphamide

Cyclophosphamide, a n lating agent. Nitrogen mustards were initially developed for use in chemical warfare during World War I to burn skin and mucous membranes. They were also the first agents successfully used as chemotherapy and remain the most frequently used chemotherapy agents.

Most nitrogen mustards, such as **cyclophosphamide**, require initial bioactivation into active metabolites by the cytochrome P (CYP) 450 enzymes in the liver. The active metabolite decomposes to form phosphamide mustard and acrolein. Phosphamide mustard produces the interstrand and intrastrand DNA crosslinks, leading to cytotoxicity.

Ifosfamide, a cyclophosphamide derivate is also metabolized by the CYP450 into its active metabolites, resulting in DNA cross linking and damage.

CLINICAL CORRELATION

The bioactivation of cyclophosphamide is an important point to keep in mind because, remember all the way back to Foundations of Pharmacology, many other drugs are inducers, inhibitors, and/ or substrates of the CYP 450 enzymes. However, many foods can be inducers or inhibitors too. A common one that interacts with many medications is grapefruit (or grapefruit juice) a potent inhibitor of CYP 3A4.

What are adverse effects of alkylating agents? (LO#2)

Cyclophosphamide can cause hemorrhagic cystitis, leading to gross hematuria, urinary frequency, dy metabolite produced during the bioactivation of nitrogen mustards, collects in the bladder and causes inflammation of the mucosal lining. **Hemorrhagic cystitis** can lead to severe blood loss and urinary obstruction due to blockage of bladder outflow by clots. The best way to prevent this is with hydration and use of chemoprotective agent, mesna. **Mesna** is activated on reaching the kidney and does not interfere with cyclophosphamide's antitumor properties. It binds to acrolein in the urine and creates an inactive compound that is safely excreted.

Alkylating agents target all rapidly dividing cells, so they will kill not only cancer cells but also other rapidly growing normal cells such as hair cells, oral and gastric mucosa, and bone marrow. Figure 3 shows **mucositis and mucosal ulceration** secondary to the toxic effects of chemotherapy. **Myelosuppression** (decreased bone marrow production of blood cells) occurs with almost all alkylating agents for this reason. Myelosuppression may lead to neutropenic fevers (fever without a high white blood cell count), bleeding, or anemia.

ACTIVE LEARNING QUESTION

What is the best way to prevent hemorrhagic cystitis caused by cyclophosphamide?

A. ○ Increase cyclophosphamide dose
B. ○ Only hydration
C. ● Hydration and mesna
D. ○ Only
E. ○ Reduce mesna dose

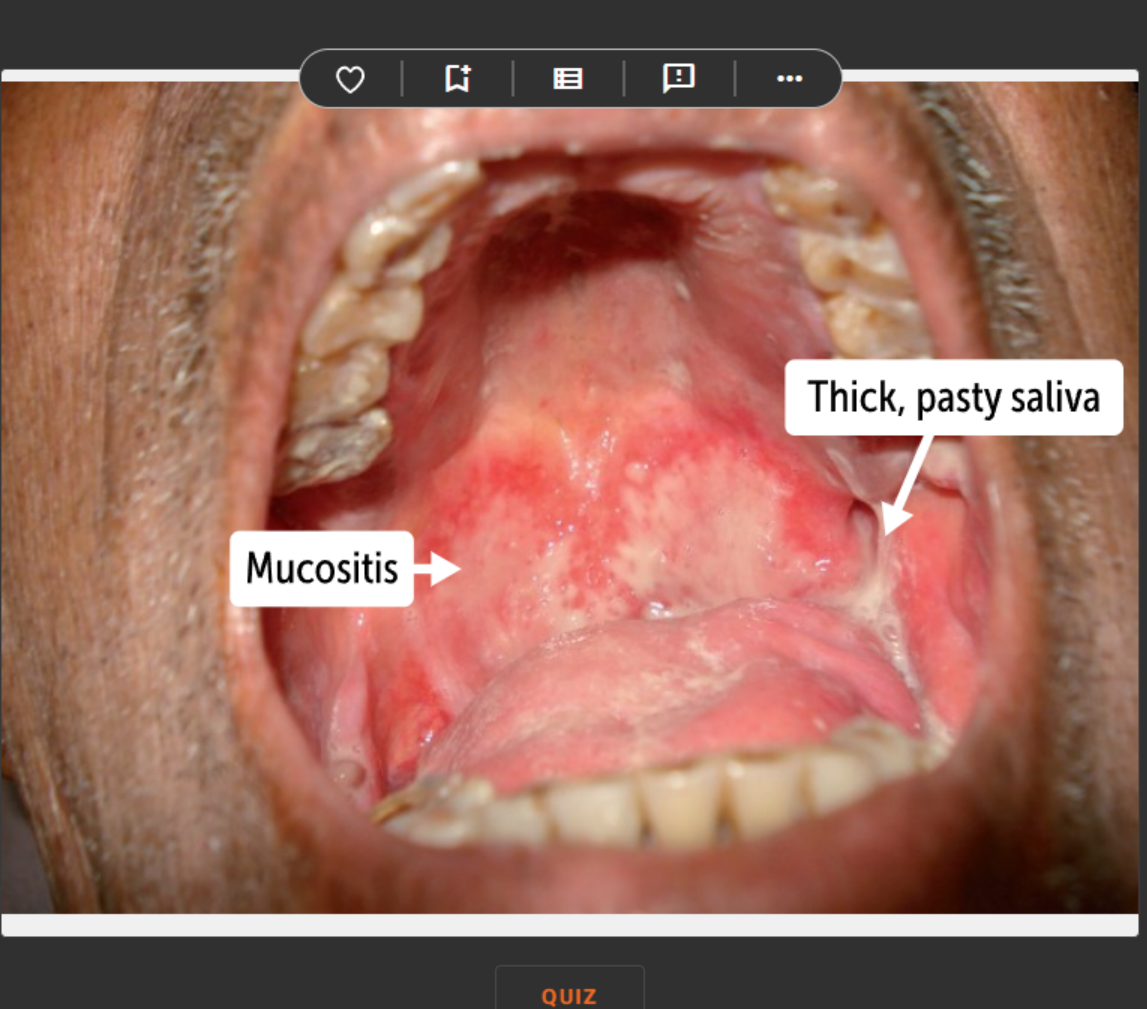
Explanation

The correct answer is hydration and mesna (C). Hydration helps to dilute the concentration of acrolein in the bladder, and mesna binds to acrolein, neutralizing its toxic effects. Increasing cyclophosphamide dose (A) would worsen the condition. Using only hydration (B) is not sufficient. Using only mesna (D) is less effective without hydration. Reducing mesna dose (E) would decrease its protective effect.

Ifosfamide use is associated with **neurotoxicity**, including encephalopathy and cerebellar dysfunctions. Methylene blue can be used as a chemoprotective agent to prevent the neurotoxicity.

CLINICAL CORRELATION

Hair follicles consist of cells with rapid turnover, which is how hair continues to grow. These cells are commonly targeted by chemotherapy agents, which is why most patients undergoing chemotherapy lose their hair.



QUIZ

Figure 3

Why are the alk ic against both rapidly growing cancers (eg, hematologic cancers) and slow-growing cancers (eg, solid tumors)?

CLINICAL CORRELATION

Like many cancer drugs, cyclophosphamide is also used to decrease the immune cell proliferation in diseases like systemic lupus erythematosus, renal glomerular disease, rheumatoid arthritis, vasculitis, and multiple sclerosis.

Carmustine

Nitrosoureas such as **carmustine** require bioactivation and then cross-link DNA. They cross-link opposite strands of DNA and RNA (interstrand cross-linking).

Adverse effects

Dose-related myelosuppression, nephrotoxicity, and hepatotoxicity can occur. Pulmonary toxicity (dose-dependent) including the development of pulmonary fibrosis and interstitial lung disease can also occur.

What is unique about the activity of nitrosoureas?

Busulfan

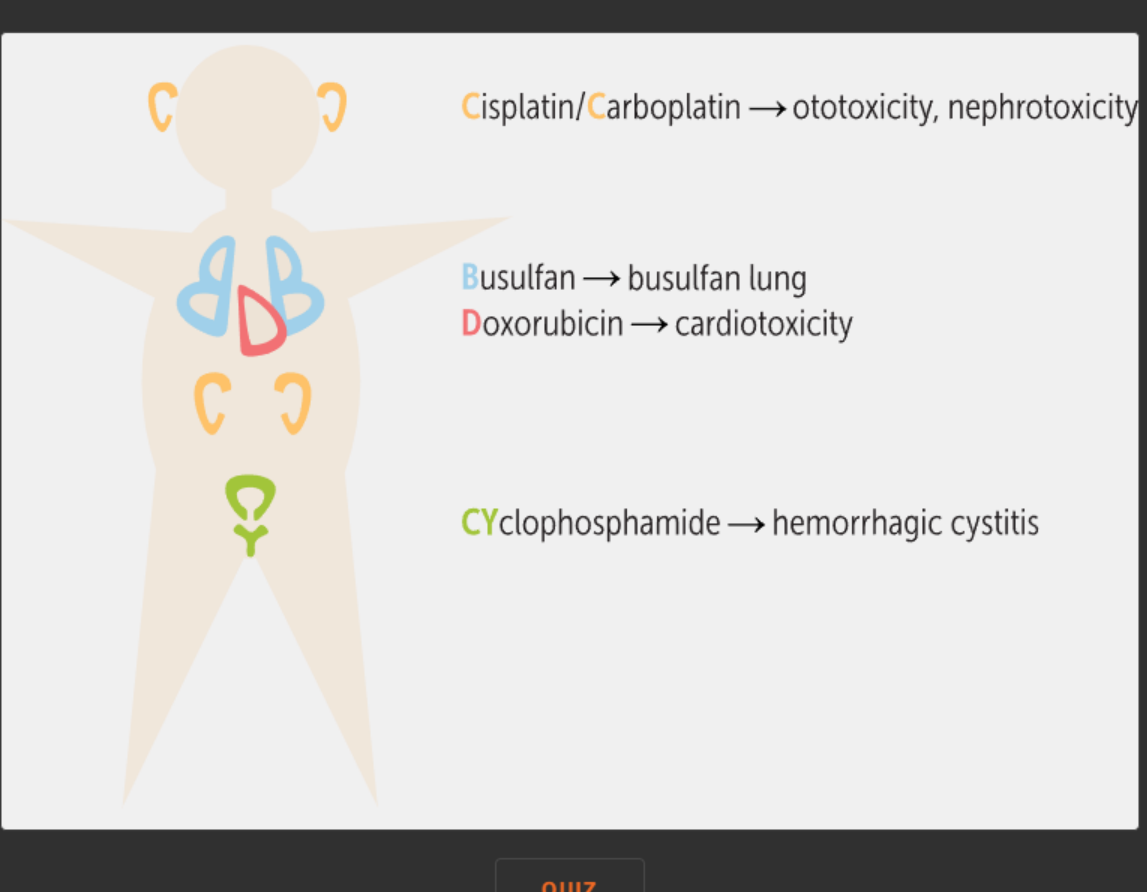
Alkyl sulfonates like **busulfan** work just like nitrogen mustards, alkylating the N-7 position of guanosine with consequent intrastrand DNA cross-linking.

Adverse effects

ADRs include myelosuppression (nearly 100% of patients) and pulmonary fibrosis. Busulfan is associated with **pulmonary toxicity**. A patient receiving multiple chemoactive drugs including busulfan who presents with cough, dyspnea, basilar crackles, or fever raises suspicion for "busulfan lung" (pulmonary fibrosis). Onset of busulfan lung can occur as early as 6 weeks after the start of treatment or up to 10 years after treatment.

CLINICAL CORRELATION

Busulfan's effect of myelosuppression has given it an interesting clinical application: children and adults who are to undergo bone marrow transplantation are given conditioning protocols with busulfan. Busulfan ablates the recipient's bone marrow and generates space for the transplantation of donor cells.


Cisplatin/Carboplatin ⟶ ototoxicity, nephrotoxicity
Busulfan ⟶ busulfan lung
Doxorubicin ⟶ cardiotoxicity
CYclophosphamide ⟶ hemorrhagic cystitis

QUIZ

Table 1 provides a summary of the alkylating and platinum agents.

Table

	Mechanism	Characteristics	Adverse Effects
Cyclophosphamide	Alkylates and cross-links DNA	Requires bioactivation by CYP450 liver enzymes	Hemorrhagic cystitis
Busulfan	Cross-links DNA (intrastrand)	Alkyl group with sulfonic acid	Pulmonary fibrosis
Carmustine	Cross-links DNA (interstrand), penetrates the CNS	Crosses blood-brain barrier, requires bioactivation by CYP450 liver enzymes	CNS toxicity
Procarbazine	Inhibits transmethylation and prevents DNA, RNA, and protein synthesis	Contains N-methyl group	Disulfiram-like reaction, serotonin syndrome
Carboplatin, cisplatin	Direct DNA damage and intrastrand cross-linking of DNA	Platinum-based molecule that is activated by hydrolyzation	Ototoxicity, nephrotoxicity

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back on MG, what can you tell her about why cyclophosphamide can cause such effects dverse effects?

You explain that cyclophosphamide is an alkylating agent that works by interfering with DNA replication. This kills the cancer cells but also affects normal cell function, leading to various side effects. Hence, the effects of hair loss and mucositis are common. A product of cyclophosphamide metabolism (acrolein) accumulates in the bladder and can cause inflammation and serious damage to the bladder lining, leading to hemorrhagic cystitis. Mesna inactivates acrolein and mitigates the risk of bladder damage. You also advise MG to drink plenty of fluids, since hydration with plenty of fluids, and paired with Mesna, can also reduce damage.

Summary

- Alkylating agents are cell cycle–independent agents that bind and cross-link DNA.
- Nitrogen mustards such as cyclophosphamide are bioactivated by CYP450 enzymes in the liver.
- Cyclophosphamide may result in hemorrhagic cystitis; administration of mesna prevents this complication.
- Alkyl sulfates such as busulfan have an affinity for myeloid cells.
- Busulfan use is often associated with pulmonary fibrosis.
- Nitrosoureas such as carmustine are able to cross the blood-brain barrier and are used to treat CNS tumors.
- The nonclassic alkylating agent procarbazine has monoamine-oxidase inhibitor activity and can cause serotonin syndrome in patients who concurrently take serotonergic drugs.

Review Questions

1. What stage of the cell cycle does an alkylating agent affect?

A. ● Cell cycle independent
B. ○ G_1 phase of cell cycle
C. ○ G_2 phase of cell cycle
D. ○ M phase of cell cycle
E. ○ S phase of cell cycle

Hide Explanation

The correct answer is cell cycle independent (A). Alkylating agents are not cell cycle–specific, making G_1 (B), G_2 (C), M (D), and S (E) incorrect. Alkylating agents can affect any part of the cell cycle, depending on the drug.

2. A patient undergoing cyclophosphamide chemotherapy comes in for follow-up. He reports changing his diet after being diagnosed with cancer and now has yogurt, a banana, and a cup of grapefruit juice for breakfast each day. The physician recommends that he discontinue the grapefruit juice because it interferes with which mechanism of cyclophosphamide?

A. ○ Alkylation
B. ● Cytochrome P450 enzyme bioactivation in the liver
C. ○ Free radical formation
D. ○ Hydrolyzation in the cell cytoplasm
E. ○ Transmethylation of methionine

Hide Explanation

The correct answer is cytochrome P450 bioactivation in the liver (B). Cyclophosphamides and nitrosoureas both require bioactivation by CYP450 enzymes into active metabolites. Alkylation at the N-7 position of guanine (A) to cross-link DNA is done by the active metabolites. Bleomycin causes free radical formation (C) to cause direct damage to DNA. Cisplatin and other platinum-based drugs are activated by hydrolyzation in the cell cytoplasm (D). Procarbazine inhibits the transmethylation of methionine (E), which inhibits protein, DNA, and RNA synthesis.

3. A 75-year-old man comes into the emergency department with shortness of breath, fever, and cough. His medical history shows he was diagnosed with chronic myeloid leukemia and started receiving a chemotherapy agent. Which of the following agents is most likely responsible for the patient's symptoms?

A. ● Busulfan
B. ○ Carmustine
C. ○ Cisplatin
D. ○ Cyclophosphamide
E. ○ Procarbazine

Hide Explanation

The correct answer is busulfan (A). Busulfan is the alkylating agent most commonly associated with pulmonary fibrosis. Patients present with fever, cough, dyspnea, and bibasilar crackles on physical examination. Carmustine (B) is a nitrosourea that causes CNS adverse effects. Platinum-based drugs such as cisplatin (C) and carboplatin are known to cause ototoxicity and nephrotoxicity. Cyclophosphamide (D) is known to cause hemorrhagic cystitis. Procarbazine (E) can cause serotonin syndrome in patients with concurrent use of serotonergic drugs.

4. What is the best way to prevent hemorrhagic cystitis caused by cyclophosphamide?

A. ○ Increasing cyclophosphamide dose
B. ○ Hydration
C. ● Hydration
D. ○ Methylene blue
E. ○ NSAIDs

Hide Explanation

The correct answer is hydration and mesna (C). Hydration helps to dilute the concentration of acrolein in the bladder, and mesna binds to acrolein, neutralizing its toxic effects. Increasing cyclophosphamide dose (A) would worsen the condition since the ADR is dose-dependent. Using only hydration (B) is not sufficient as the toxic metabolite remains active. Methylene blue (D) is used to treat ifosfamide toxicity. NSAIDs (E) are nephrotoxic and could worsen the hemorrhagic cystitis would decrease its protective effect.

How would you rate this Brick?

👍 👎

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ Explain nd alkylating agents.

The correct answer is hydration and mesna (C). Hydration helps to dilute the concentration of acrolein in the bladder, and mesna binds to acrolein, neutralizing its toxic effects. Increasing cyclophosphamide dose (A) would worsen the condition since the ADR is dose-dependent. Using only hydration (B) is not sufficient as the toxic metabolite remains active. Methylene blue (D) is used to treat ifosfamide toxicity. NSAIDs (E) are nephrotoxic and could worsen the hemorrhagic cystitis would decrease its protective effect.

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Explain the mechanism of action of commonly used alkylating agents.~~

02 ☑ ~~Explain the adverse effects of alkylating agents and strategies for mitigating these effects.~~

Objective Confidence: 2/2